SDC2026_ICEMAN_TechDeclaration

1. Team and project overview

Field	Response
Team Name	ICEMAN
Project / Solution Name	Future-Ready RPG
Name (Leader)	Eashwar Singh Sidhu
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University Affiliation(s)	Asia Pacific University
Live Prototype URL	https://sdggez.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/eaergfyl-maker/SDGGEZ

2. Architecture and technology stack

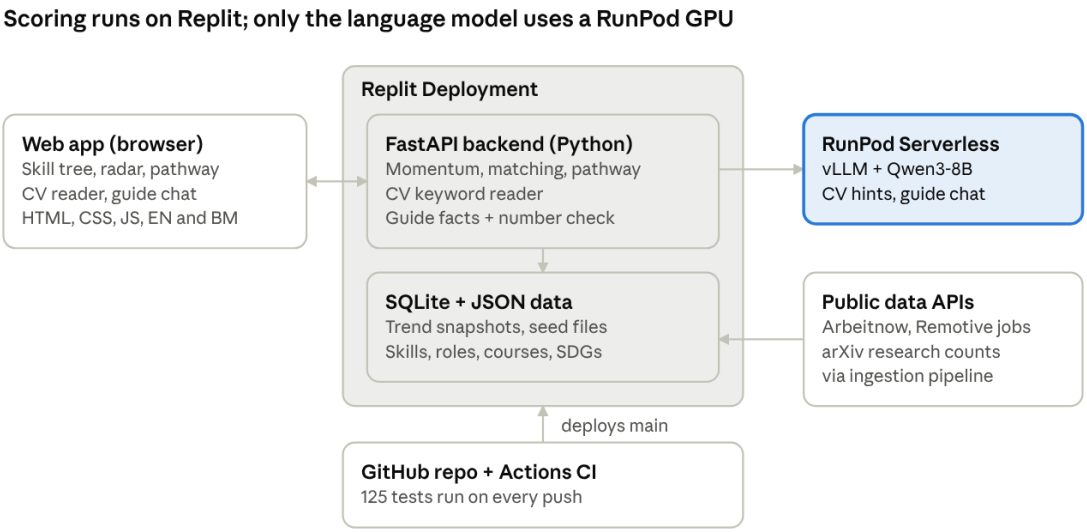
2.1 System architecture overview

The user works in a browser app (HTML, CSS and JavaScript, no build step) served by one FastAPI backend on Replit Deployments. The backend does all scoring in plain Python: skill momentum, role matching, pathways, SDG tags and the keyword CV reader. It reads trend snapshots from SQLite, falling back to snapshot files committed in the repository, plus JSON data for skills, roles, courses and SDGs.

An ingestion pipeline pulls job postings from the Arbeitnow and Remotive APIs and paper counts from the arXiv API, and saves each run as a dated snapshot.

The only GPU work is the language model. Qwen3-8B runs on a RunPod Serverless endpoint (vLLM). The backend calls it for two features: extra CV skill suggestions and "Ask your guide" answers. It sends only redacted CV text or facts the app computed, then checks every reply in code before showing it. If RunPod is unavailable, both features hide and the rest of the app works.

Code lives on GitHub; GitHub Actions runs 125 tests on every push, and Replit deploys the main branch. Keys are held in Replit Secrets.



System architecture · Replit app, RunPod model, public data APIs

Word count: 183 (limit 200).

2.2 Languages, frameworks and libraries

Versions are those installed and tested on 24 October 2026; `requirements.txt` sets the minimums.

Name & version	Purpose in project	License
Python 3.12 (Replit), 3.11 (CI)	Backend language	PSF License
HTML, CSS, JavaScript (ES modules)	Frontend, no framework or build step	Open web standards
FastAPI 0.159.1	Web API framework	MIT
Starlette 1.9.0	Web toolkit under FastAPI; static files	BSD-3-Clause
Uvicorn 0.54.0	Web server	BSD-3-Clause
Pydantic 2.14.1	Request validation	MIT
httpx 0.28.1	Calls to job, arXiv and RunPod APIs	BSD-3-Clause
feedparser 6.0.14	Reads arXiv API responses	BSD-2-Clause
PyYAML 6.0.3	Story files	MIT
APScheduler 3.11.3	Installed from requirements.txt; not used by the current code (refresh runs on request)	MIT
jsonschema 4.26.0	Validates the data files	MIT
SQLite (Python standard library)	Trend snapshot storage	Public domain

Name & version	Purpose in project	License
pytest 9.1.1	Automated tests	MIT
vLLM 0.32.0 in RunPod worker-vllm v2.29.0	Serves Qwen3-8B on RunPod	Apache 2.0 (vLLM); MIT (worker-vllm)

3. APIs and cloud services

3.1 External APIs

No API keys or secrets appear in this document or the repository.

API and provider	Purpose in project	Access tier
Job Board API, Arbeitnow	Main job-posting feed: share of postings mentioning each skill, last 5 days vs the 5 before	Free, public, no key
Remote Jobs API, Remotive	Supplementary job postings (about 10 per call)	Free, public, no key; Remotive named and linked
arXiv API, Cornell University	Monthly research-paper counts for 12 AI and technology skills	Free, public, no key
RunPod Serverless API (`/run`, `/status`), RunPod	Sends redacted CV text or computed facts to our Qwen3-8B endpoint	Sponsor-provided credit (US$25)

Course pages and UN SDG pages are linked for users to open; the app does not call them.

3.2 Cloud platforms and deployment

The prototype is deployed on Replit Deployments from the main branch; the language model is a separate RunPod Serverless endpoint that scales to zero when idle.

Service & provider	Purpose in project	Plan / region
Replit Workspace and Deployments	Shared development and the live app (sdggez.replit.app); keys in Replit Secrets	Replit Core (sponsor-provided); region (Reserved VM, 0.5 vCPU / 2 GiB RAM; Asia region)
RunPod Serverless, vLLM worker	Runs Qwen3-8B for CV suggestions and the guide chat	Pay per second from sponsor credit; 24 GB GPU (RTX 4090 observed), 48 GB fallback; US (US-NC-1 observed); max 1 worker, 0 always-on
GitHub and GitHub Actions	Source code, pull requests, 125 automated tests on every push	Free plan

4. AI tools, models and datasets

4.1 AI models used in the solution

One language model is embedded in the app; all scoring, matching and pathway logic is deterministic Python, not a model.

Model & version	Source / provider	Usage	License / terms
Qwen3-8B (Qwen/Qwen3-8B), thinking mode off, temperature 0, 8,192-token context	Qwen team, Alibaba Cloud, via Hugging Face; self-hosted on our RunPod endpoint	Pre-trained, used as is (no fine-tuning). (1) Suggests CV skills from a fixed list of 39, each with a quote from the CV. (2) Answers "Ask your guide" questions from facts the app computed	Apache 2.0

In both uses the code checks the output: a suggested skill must exist in our list and its quote must appear word for word in the CV; a guide answer that contains a number not found in the user's data is withheld.

4.2 Datasets

No dataset contains real personal data; the people and CVs in the demo are invented.

Dataset name	Source / URL	Type	License	Personal data?
Job-demand snapshots (22 and 24 Oct 2026)	Arbeitnow API, Remotive API; data/snapshot_seed.json , data/snapshot_previous.json	Public (aggregated counts per skill)	API terms of each provider	N
Research-paper counts	arXiv API	Public (monthly counts)	arXiv API Terms of Use	N
O*NET 31.0 Database	onetcenter.org	Public (occupation reference for our 16 roles)	CC BY 4.0, credited and adapted	N
UN SDG targets	sdgs.un.org/goals	Public	UN terms of use	N
Skills, roles and SDG map (39 skills, 16 roles)	data/skills.json, data/roles.json, data/sdg_map.json	Self-created from the sources above	Apache 2.0 (our repository)	N
Course catalogue (10 verified courses)	data/courses.json, each linked to the provider's public page	Self-collected, each checked by hand with a date	Apache 2.0 (our list); course content belongs to providers	N

Dataset name	Source / URL	Type	License	Personal data?
Starting classes (4 sample workers)	data/personas.json	Synthetic	Apache 2.0	N
Sample CVs (4)	data/sample_cvs/	Synthetic, labelled "invented, not a real person"	Apache 2.0	N

4.3 AI-assisted development tools

Did the team use generative-AI tools to help create project material? Yes. Each use is logged with its date and scope in docs/ai-tools-log.md.

Tool & version	Used for	Extent of use
Claude Code (Claude Opus 5.5; Claude Sonnet 5.5), Anthropic	Backend and frontend code (scoring, CV reader, AI guide, data pipeline fixes, UI), tests, data files, documentation, and drafting this declaration and the proposal	Major
Replit Agent	Replit environment setup (.replit, replit.md) and some frontend code	Moderate
Claude chat on claude.ai (Claude Sonnet 5.5), Anthropic	Guidance on git, GitHub, Replit and deployment; drafted the refresh token and cooldown, its tests, the dependency minimums and the Replit port setting	Moderate

How we reviewed, tested and took responsibility for AI-generated output. A team member asked for every AI change, read the diff and approved it before it reached main, either directly or through a pull request. Automated tests run on every push, and each feature was also checked by hand in the browser and against invented sample data. Facts and statistics were checked against their sources, and courses were verified by opening each page. We can explain every part of the code.

Word count: 82 (limit 100).

5. Sponsor-provided resources

Did the project use any official SDC sponsor-provided resource? Yes.

Sponsor	Resource	How it was used in the project
Replit	Platform: Replit Core for each participant	Shared workspace, Replit Agent for environment setup, Secrets for keys, and Replit Deployments for the live prototype
RunPod	Credits: US$25 for the team	Serverless GPU endpoint running Qwen3-8B for CV skill suggestions and the guide chat
Adam Alias bin Amin	Zoom: Thu 22 Oct	Asked if the game design was unique

Essential or replaceable? Replit is essential as our host and the required platform, though the app is a standard FastAPI app that runs anywhere Python does. RunPod is essential for the two AI features, because the handbook requires GPU work on RunPod, but not for the core journey: profile, radar, matching, pathway, SDGs and scenarios run without it, and the AI features hide when the endpoint is off. Another GPU host could serve the same open model.

Word count: 77 (limit 100).

6. Security, privacy and responsible use

6.1 Data handling and privacy

No user data is stored on our server. There are no accounts: a nickname, ticked skills and progress are saved only in the user's own browser, and the character sheet can download or delete them.

Pasted or uploaded CV text is processed in memory and never stored or logged. Emails and numbers of nine or more digits (phone and IC numbers) are removed before any matching. Only that redacted text, or facts the app computed, is sent to our RunPod endpoint; RunPod handles requests under its own terms. Guide questions and answers are not stored or logged by our app.

The app asks users to use invented CVs, and the demo uses invented samples only. API keys stay in Replit Secrets and are never sent to the browser.

Word count: 128 (limit 150).

6.2 Known limitations and risks

- **Data coverage.** Job postings are mostly German and European with about one week of history, so figures swing between snapshots. We drop figures based on too few postings or papers, label every number with its source and date, and show scenarios as ranges.
- **Model errors.** An 8B model can misread or overstate. CV suggestions must quote the CV and stay unticked until confirmed; guide answers with unsupported numbers are withheld.
- **Single provider and cold starts.** If RunPod is slow or down, the AI features hide and the core journey still works; a first request can take over a minute.
- **Gaps.** Verified courses cover 8 of 39 skills; the keyword CV reader works in English only; the Malay interface text was AI-drafted and needs a fluent review.
- **Fairness.** Scoring uses skills only, never name, age, gender or background, and a test checks that identical skills give identical results.

Word count: 147 (limit 150).

7. Compliance and readiness

Ticked boxes were checked against the repository on 24 October 2026; each open box has a note on what is left.

- ☒All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒The prototype described here is accessible to judges through the links provided in Section 1.
- ☒All fields in this template have been completed within the word limits stated. 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature	 [Eashwar Singh Sidhu,]
Date	24/10/2026